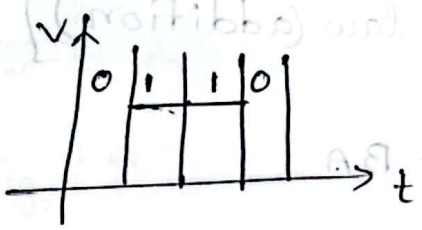
1. (a)

Binary digits (Bits) : Binary digits are 0 and 1 which are the digits in Binary system.

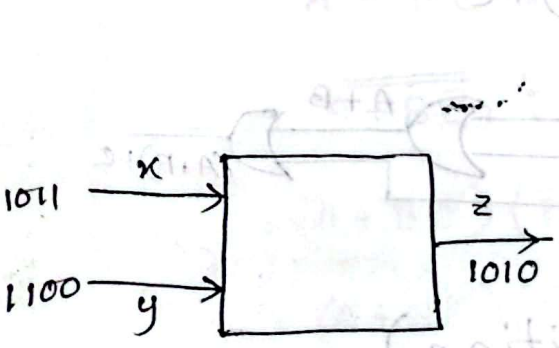
0 (low or off)

1 (high or on)

Logical Levels : The voltages used to represent a '1' and a '0' are called logic levels.



Digital waveforms : A digital waveform is a graphical representation of how a signal varies over time between logical levels.



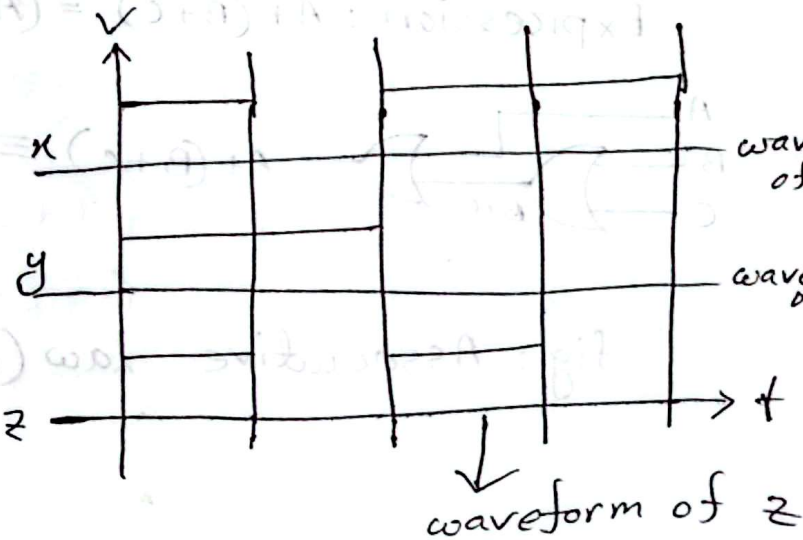


fig : Digital waveform

1.(b)

The basic laws of boolean algebra :

1. Commutative laws :

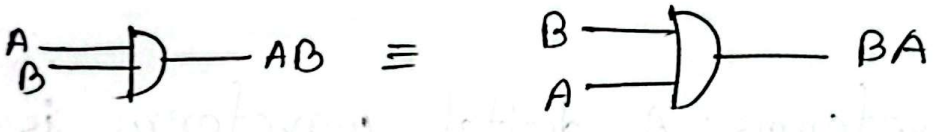
→ for addition and multiplication.

Expression : $A + B = B + A$



[fig of commutative law (addition)]

Expression : $AB = BA$



[fig of commutative law (multiplication)]

2. Associative law :

Expression : $A + (B + C) = (A + B) + C$

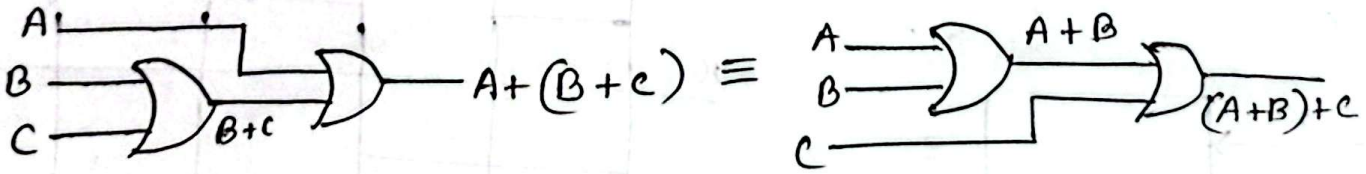


fig : Associative law (Addition)

Expression: $A(BC) = (AB)C$

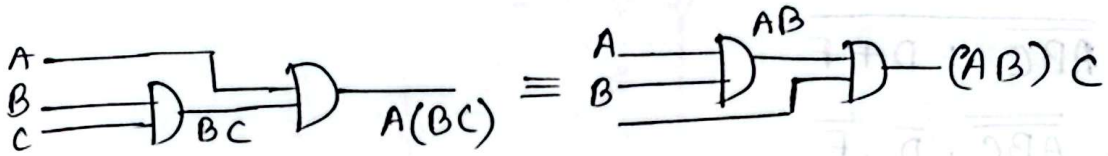


Fig: Associative law (multiplication)

3. Distributive law:

$$A(B+C) = AB + AC$$

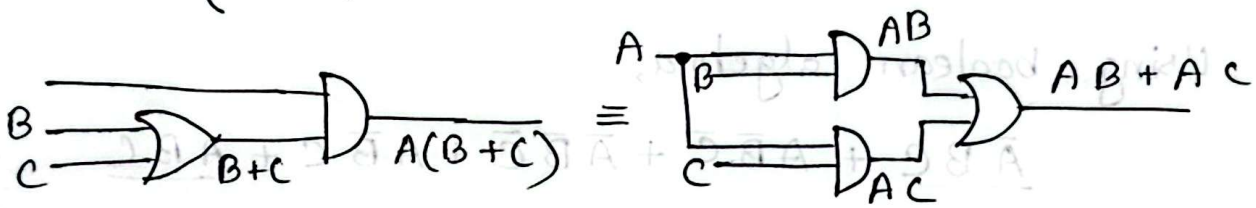


Fig: Distributive law

1.(c)

i) Applying DeMorgan's Theorem,

$$\overline{\overline{A + B\overline{C} + D(E+\overline{F})}}$$

$$= \overline{\overline{A + B\overline{C}}} \cdot \overline{\overline{D(E+\overline{F})}}$$

$$= (A + B\overline{C})(\overline{D} + \overline{\overline{E+\overline{F}}})$$

$$= (A + B\overline{C})(\overline{D} + E + \overline{F})$$

ii) Applying De Morgan's Theorem,

$$\overline{\overline{ABC} + D + F}$$

$$= \overline{\overline{ABC}} \cdot \overline{D} \cdot \overline{F}$$

$$= ABC\,\overline{D}\,\overline{F}$$

2.(a)

Using boolean algebra,

$$\underline{\bar{A}BC} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + \underline{AB\bar{C}}$$

$$= BC(A+\bar{A}) + A\bar{B}(C+\bar{C}) + \bar{A}\bar{B}\bar{C} \quad [A+\bar{A}=1]$$

$$= BC\cdot 1 + A\bar{B}\cdot 1 + \bar{A}\bar{B}\bar{C}$$

$$= BC + \bar{B}(A+\bar{A}\bar{C})$$

$$= BC + \bar{B}(A+\bar{C}) \quad [A+\bar{A}B = A+B]$$

$$= BC + \bar{B}A + \bar{B}\bar{C}$$

2.(b)

POS ⇒ POSITIVE 0
SOP ⇒ NEGATIVE 0

POS to SOP:

$$Y : (\bar{A}+B+C)(\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})(A+B+C)$$

: 1 0 0 | 1 1 0 | 0 1 1 | 0 0 0

↳ Output zero (0)

Absent : ∅1, 2, 5, 7

Truth table :

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

SOP expression

~~$A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$~~

$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + ABC$

2. (c)

Given SOP expression :

$A\bar{B}C + \bar{A}B + AB\bar{C} + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + ABCD$

1 0 0 0 0 1 0 0 1 1 0 0 ~~0 1 0 1~~ ~~1 0 1 0~~ 1 1 1 1

1 0 1 1 0 1 0 1 1 1 0 1

0 1 1 0

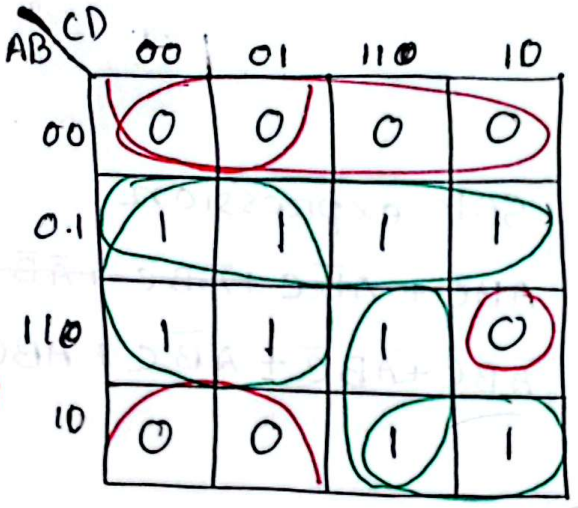
0 1 1 1

The SSOP expression is :

$A\bar{B}CD + A\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD$

$+ AB\bar{C}D + AB\bar{C}\bar{D}$ ~~$+ \bar{A}B\bar{C}D + A\bar{B}C\bar{D}$~~ $+ ABCD$

The expression is mapped on Karnaugh map as follows :



Minimized, $Y = B\bar{C} + \bar{A}B + ACD + A\bar{B}C$

SPOS Expression :

$(A+B+C+D)(A+B+C+\bar{D})(A+B+\bar{C}+\bar{D})(A+B+\bar{C}+D)$

$(\bar{A}+\bar{B}+\bar{C}+D)(\bar{A}+B+C+D)(\bar{A}+B+C+\bar{D})$

Minimized, $Y = (A+B)(B+C)(\bar{A}+\bar{B}+\bar{C}+D)$

6. Given SOP expression :

$$\bar{B}C + A\bar{B} + A\bar{B}C + A\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D}$$

0 0 1 0 | 1 0 0 0 | 1 0 1 0 | 1 0 1 0 | 0 0 1 1 | 1 0 0 0
0 0 1 1 | 1 0 0 1 | 1 0 1 1
1 0 1 0 | 1 0 1 0
1 0 1 1 | 1 0 1 1

The standard SOP expression:

$$\bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$$

The expression is mapped on K-map as follows:

AB \ CD	00	01	11	10
00			1	1
01				
11				
10	1	1	1	1

Minimized expression:

$$A\bar{B} + \bar{B}C$$

8.a) $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$

$= BC(A+\bar{A}) + \bar{B}\bar{C}(A+\bar{A}) + A\bar{B}C$

$= BC + \bar{B}\bar{C} + A\bar{B}C$

~~$= BC + \bar{B}$~~ $= C(B + A\bar{B}) + \bar{B}\bar{C}$ $[A + A\bar{B} = A + B]$

$= C(B+A) + \bar{B}\bar{C}$

$= BC + AC + \bar{B}\bar{C}$

8(b)

$$A\bar{B} + A\bar{B}\bar{C} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$$

$$= A\bar{B}(1+\bar{C}) + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$$

$$= A\bar{B} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$$

$$= A\bar{B}(1+\bar{C}\bar{D} + \bar{C}\bar{D}E)$$

$$= A\bar{B}$$

7/ $X = (\bar{A}+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(A+B+C)$

SPOS

Truth table for given SPOS:

A	B	C	X	Sum Term
0	0	0	0	$A+B+C$
0	0	1	1	
0	1	0	0	$A+\bar{B}+C$
0	1	1	1	
1	0	0	0	$\bar{A}+B+C$
1	0	1	0	$\bar{A}+B+\bar{C}$
1	1	0	1	
1	1	1	1	

9) DeMorgan

i) $\overline{A + B\bar{C} + D(\bar{E} + \bar{F})}$

$= \overline{A + B\bar{C}} \cdot \overline{D(\bar{E} + \bar{F})}$

$= (A + B\bar{C})(\bar{D} + \overline{\bar{E} + \bar{F}})$

$= (A + B\bar{C})(\bar{D} + E + \bar{F})$

ii) $\overline{\overline{ABC} + D + F}$

$= \overline{\overline{ABC}} \cdot \overline{D + F}$

$= ABC\,D'F'$

10.a) $(A+B)(B+C)$ Boolean expression to POS

$= (A + B + C\bar{C})(A\bar{A} + B + C)$

$= (A+B+C)(A+B+\bar{C})(A+B+C)(\bar{A}+B+C)$

$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)$

b) $\bar{A}BC + \bar{A}B\bar{C} + A\bar{B}C + ABC$ (SSOP to SPOS)

$\quad 011 \quad 010 \quad 101 \quad 111$

The missing terms among the $2^3 = 8$ terms are 000, 001, 100, 110

So, the required SPOS expression:

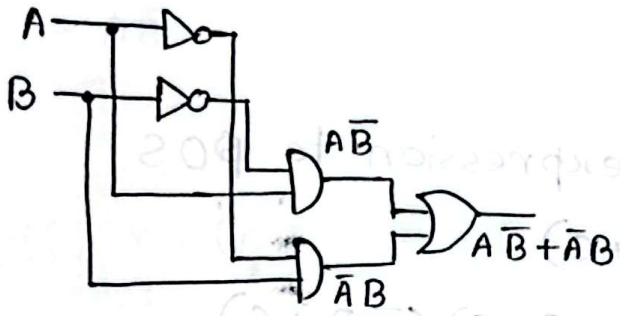
$(A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$

2. XOR :

$X = A \oplus B$

$= A\bar{B} + \bar{A}B$

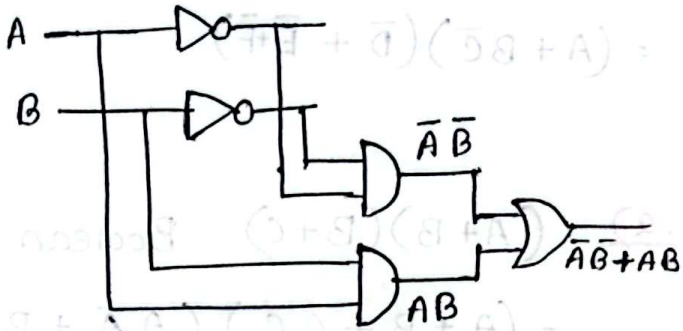
Circuit Diagram:



XNOR :

$X = \overline{A \oplus B}$

$= \bar{A}\bar{B} + AB$



4. i) $A\bar{B} + A(\overline{B+C}) + B(\overline{B+C})$

$= A\bar{B} + A\bar{B}\bar{C} + B\bar{B}\bar{C}$

$= A\bar{B}(1+\bar{C}) + 0$

$= A\bar{B}$

(Proved)

ii) $(A+B)(A+C)$

$= AA + AC + AB + BC$

$= A + AC + AB + BC$

$= A(1+C+B) + BC$

$= A + BC$

(Proved)

iii) $AB + A(B+C) + B(B+C)$:

$= AB + AB + AC + BB + BC$

$= AB + B + BC + AC$

$= B(A+1+C) + AC$

$= B + AC$

(Proved)

5) i) $\overline{(\overline{A+B+C})\,\overline{D}}$

$= \overline{\overline{A+B+C}} + \overline{\overline{D}}$

$= A+B+C+D$

ii) $(\overline{\overline{AB} + (\overline{CD+EF})})$

$= \overline{\overline{AB}} \cdot (\overline{\overline{CD+EF}})$

$= AB(CD+EF)$

6. Given SOP Expression:

Neg 0

$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$

$+ AB\bar{C}\bar{D} + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + ABCD$

The expression is mapped on K-map as follows:

AB \ CD	00	01	11	10
00	1	1	1	
01		1	1	
11	1		1	1
10	1	1		1

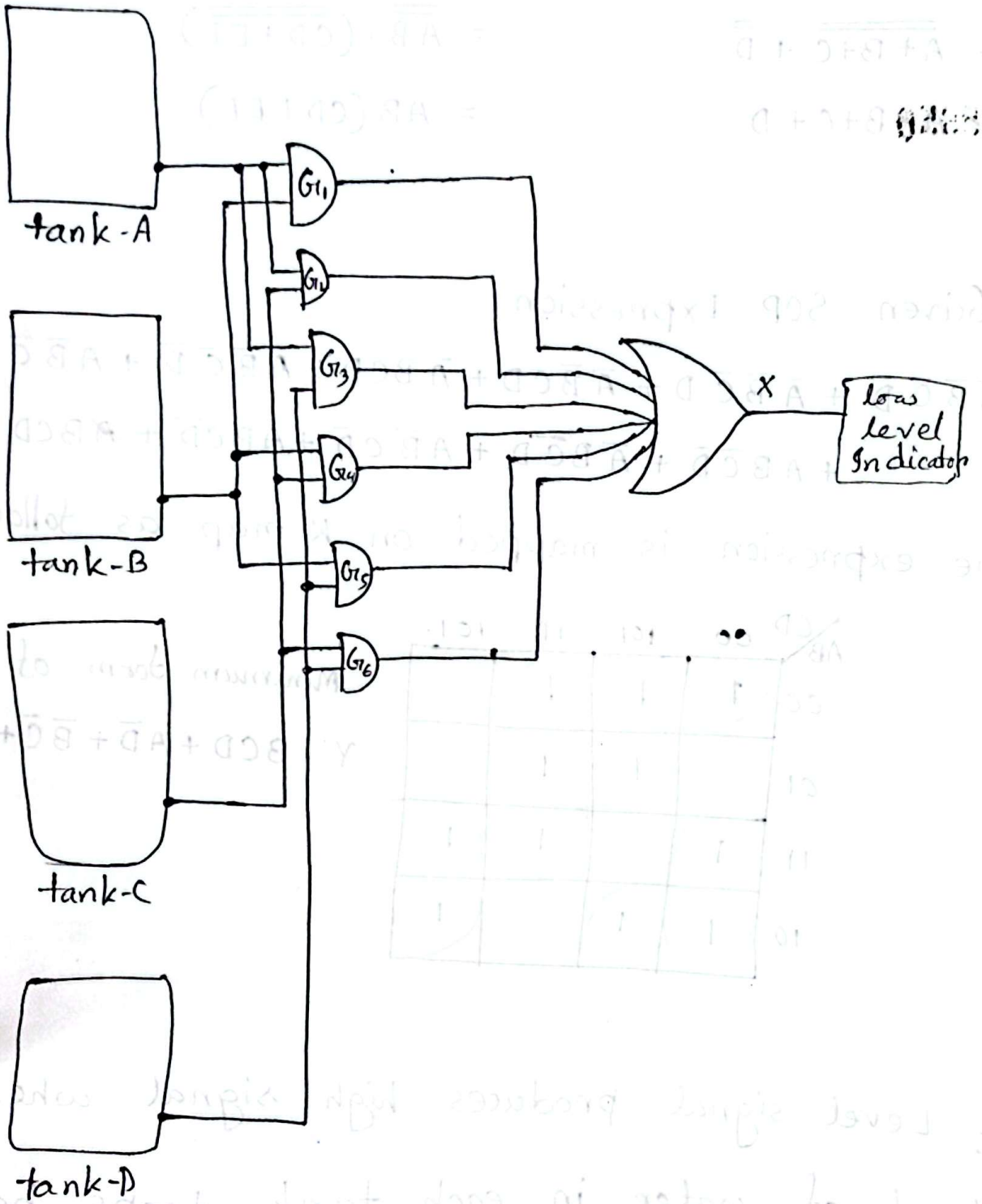
Minimum form of SOP:

$Y = BCD + A\bar{D} + \bar{B}\bar{C} + \bar{A}D$

3. Level signal produces high signal when level of water in each tank drops below a specified point.

Total number of tank = 4

3


tank-A
tank-B
tank-C
tank-D
G1
G2
G3
G4
G5
G6
X
Low level Indicator

2.a) i) DeMorgan

$$\overline{\overline{A+B\bar{C}}+D(\bar{E}+\bar{F})}$$

$$= \overline{\overline{A+B\bar{C}}} \cdot \overline{D(\bar{E}+\bar{F})}$$

$$= (A+B\bar{C})\,\bar{D} + (\bar{E}+\bar{F})$$

$$= (A+B\bar{C})(\bar{D}+E+\bar{F})$$

ii) $\overline{\overline{ABC}+D+F}$

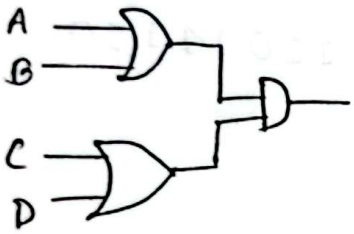
$$= \overline{\overline{ABC}} \cdot \overline{D+F}$$

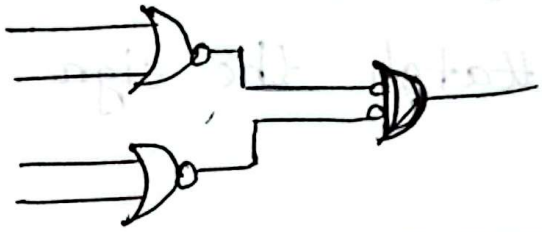
$$= ABC\,\bar{D}\,\bar{F}$$

ii) $X = (A+B)(C+D)$

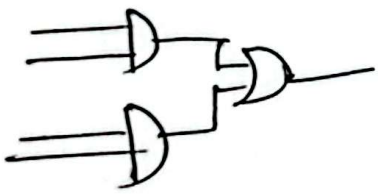
$$\overline{(A+B)+(C+D)}$$

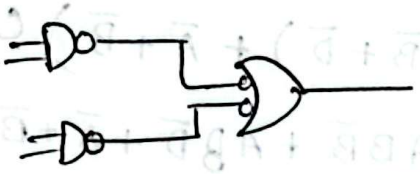
$$= (A+B)\cdot(C+D)$$





$X = AB + CD$





NAND NOT-OR

NOR NOT AND

NAND AND POS

SOP NOT OR

b) Multiplicand → 01011011 (+ve)

Multiplier → 11111010 (-ve).

Take the 2's complement of the multiplier as the sign bit is 1 denoting negative bit.

00000110

```
  01011011
  00000110
  --------
  00000000
  01011011 0
  ----------
  01011011 0
 0101 101100
 -----------
 1000100010
```

Sign bit of the product is 1, take the 2's complement of the product,

```
  01110111 01
            +1
  ----------
  01110111 10
```

Now, attatch the sign bit 1 1110111 10

c) i) $(AB(C+\overline{BD})+\overline{AB})C$

$= (AB(C+\overline{B}+\overline{D})+\overline{A}+\overline{B})C$

$= (ABC+AB\overline{B}+AB\overline{D}+\overline{A}+\overline{B})C$

$= ABC+ABC\overline{D}+\overline{A}C+\overline{B}C$

$= ABC(1+\overline{D})+\overline{A}C+\overline{B}C$

$= ABC+\overline{A}C+\overline{B}C$

ii) $ABC[(AB+\overline{C})(BC+AC)]$

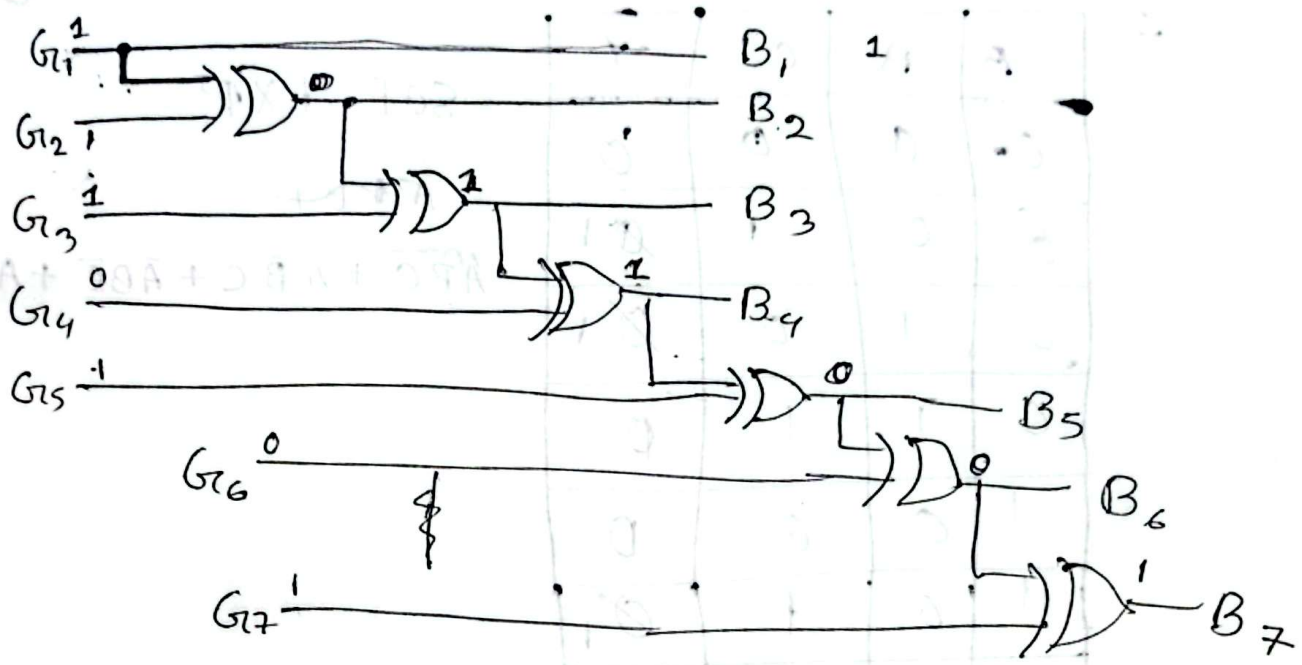
$= ABC[ABC+ABC+BC\overline{C}+AC\overline{C}]$

$= ABC \cdot ABC$

$= ABC$

d) i) Gircay : $G_1\ G_2\ G_3\ G_4\ G_5\ G_6\ G_7$
1 1 1 . 0 1 0 1

Binary : 1 0 1 1 0 0 1
$B_1\ B_2\ B_3\ B_4\ B_5\ B_6\ B_7$



ii) 7952 → BCD : 0111 1001 0101 0010

278.3 → BCD : 0010 0111 1000 0011

10735

1010 0000 1101 0101 (1)
5
0110 0110 0110
10000 0111 0011 0101

Carry or over 9 হলে

add 6 (0110)

all

4.① POS to SOP

$(\bar{A}+B+C)(\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})(A+B+C)$

1 0 0 1 1 0 0 1 1 0 0 0

Absent : ~~000, 001, 010, 101~~ ~~111, 100, 110, 011~~ 111, 001, 010, 101

∴

A	B	C	Y
0	0	0	0
0	0	1	Ø 1
0	1	0	Ø 1
0	1	1	0
1	0	0	0
1	0	1	Ø 1
1	1	0	0
1	1	1	1

SOP EXP :

~~$\bar{A}+\bar{B}+$~~

$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C$

Neg

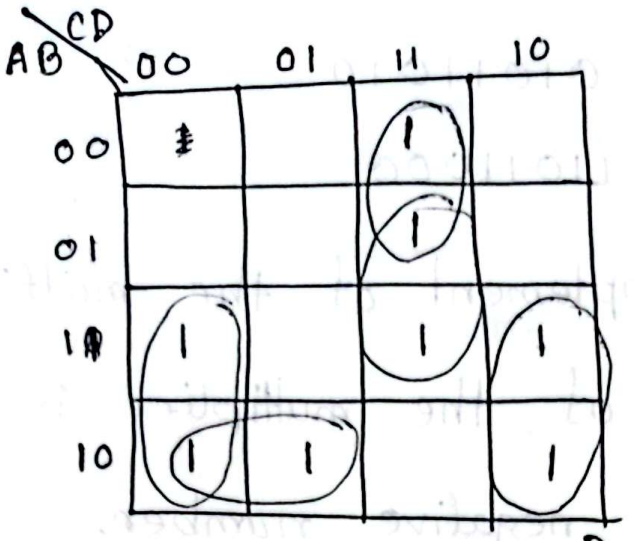
ii) Given SOP expression :

Neg

$Y = \bar{A}BCD + \bar{A}\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D}$

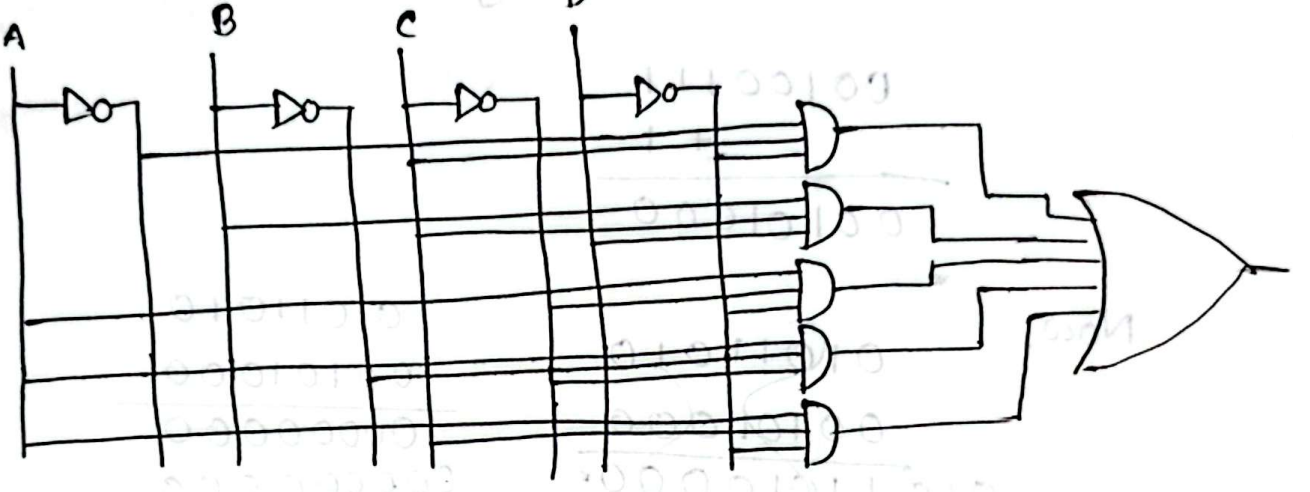
$+ ABC\bar{D} + AB\bar{C}\bar{D} + ABCD$

Karnaugh map of this expression :



Minimized:

$Y = \bar{A}CD + BCD + A\bar{C}\bar{D} + A\bar{B}\bar{C} + AC\bar{D}$



neg 0

(b) Given SOP expression:

$\bar{A}BCD + AB\bar{C} + ABCD + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}D + AB\bar{C}\bar{D}$

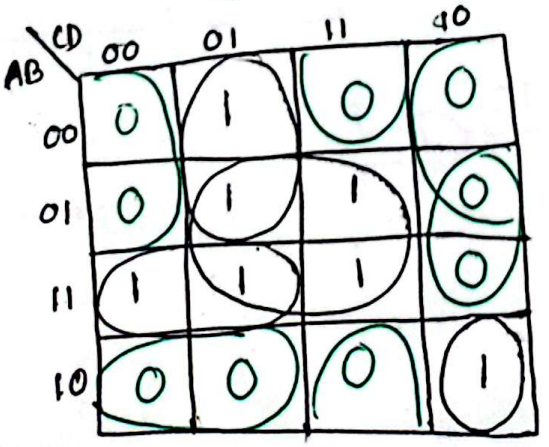
0111 1100 1111 1010 0001 0101 1100

1101

minimum

POS: $(A+D)(\bar{A}+B+C)(B+\bar{C}+\bar{D})(\bar{B}+\bar{C}+D)$

Karnaugh map:



SPOS: $(A+B+C+D)(A+B+\bar{C}+\bar{D})(A+B+\bar{C}+D)(A+\bar{B}+C+D)(A+\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+\bar{C}+D)(\bar{A}+B+C+D)(\bar{A}+B+\bar{C}+\bar{D})$

2. b) Multiplicand : 01011010

Multiplier : 11011000

Take the 2's complement of the multiplier as the sign bit of the multiplier is Ø 1 which indicates a negative number.

```
  00100111
+        1
----------
  00101000
```

Now;

```
  01011010
  00101000
----------
 0101101000000
 0000000000000
 0101101000000
----------
 0110010101000
```

```
  01011010
  00101000
----------
  00000000
 000000000
----------
 0000000000
 0000000000
 01011010000
----------
    0000
 0101101000000
----------
 0111000010000
```

Take the 2's complement of the result,

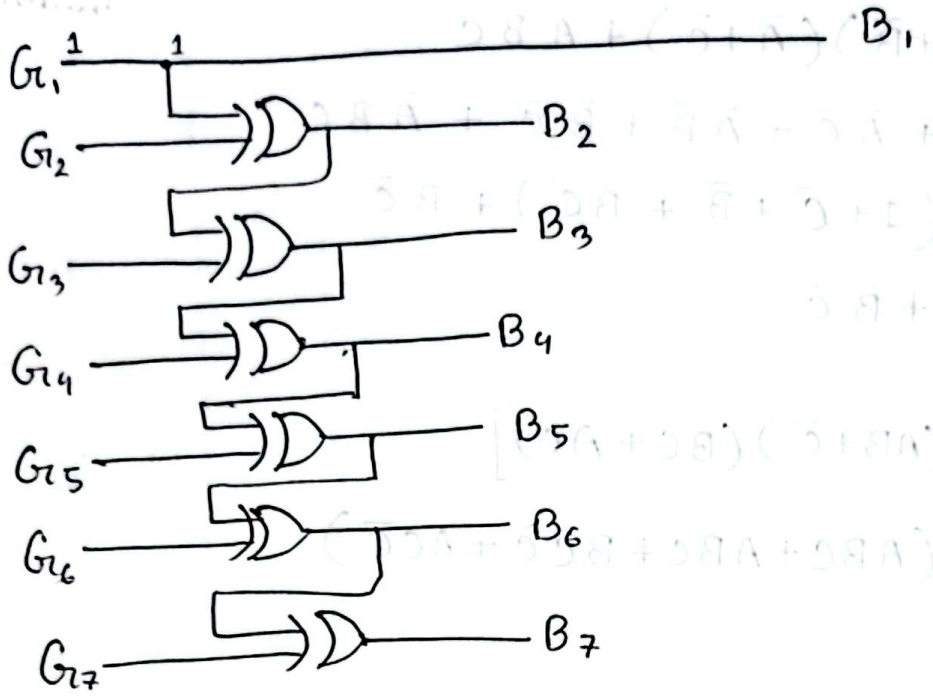
```
 1000111101111
+             1
----------------
 1000111110000
```

(c) Gray, G :

G_1	G_2	G_3	G_4	G_5	G_6	G_7
1	0	1	0	1	1	1

Binary, B :

B_1	B_2	B_3	B_4	B_5	B_6	B_7
1	1	0	0	1	0	1

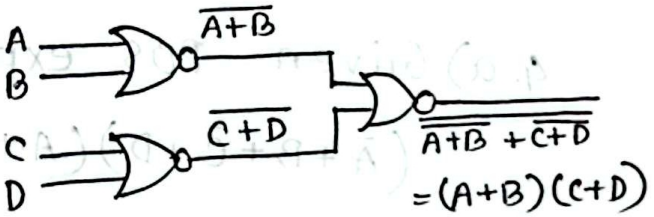


d) i) Implement eqn using NOR

$$X = (A+B)(C+D)$$

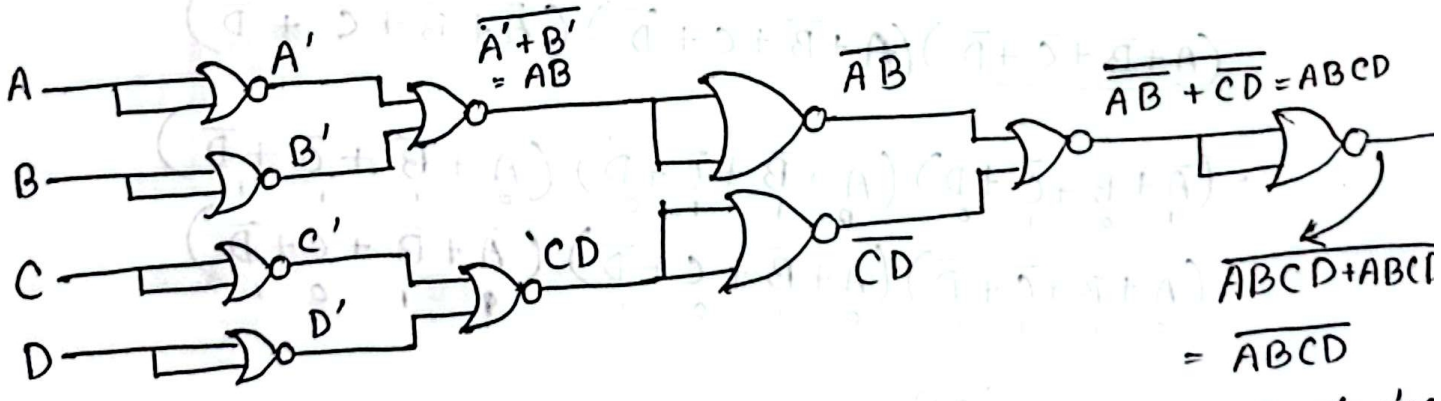
$$= \overline{(\overline{A+B}) + (\overline{C+D})}$$

$$= A$$



ii) $\overline{AB} + \overline{CD}$

$$= \bar{A} + \bar{B} + \bar{C} + \bar{D}$$



Simplification

3.a) i) $\overline{AB+AC}+\bar{A}\,\bar{B}\,C$

$= \overline{AB}\cdot\overline{AC}+\bar{A}\,\bar{B}\,C$

$= (\bar{A}+\bar{B})(\bar{A}+\bar{C})+\bar{A}\,\bar{B}\,C$

$= \bar{A}+\bar{A}\,\bar{C}+\bar{A}\,\bar{B}+\bar{B}\,\bar{C}+\bar{A}\,\bar{B}\,\bar{C}$

$= \bar{A}(1+\bar{C}+\bar{B}+\bar{B}\bar{C})+\bar{B}\,\bar{C}$

$= \bar{A}+\bar{B}\,\bar{C}$

ii) $ABC\left[(AB+\bar{C})(BC+AC)\right]$

$= ABC\,(ABC+ABC+BC\bar{C}+AC\bar{C})$

$= ABC$

4. a) Given POS expression:

$(\bar{A}+B+\bar{C}+D)(A+\bar{B}+\bar{C})(A+\bar{C}+\bar{D})(\bar{B}+C+\bar{D})$

$=(\bar{A}+B+\bar{C}+D)(A+\bar{B}+\bar{C}+D\bar{D})(A+B\bar{B}+\bar{C}+\bar{D})$

$(A\bar{A}+\bar{B}+C+\bar{D})$

$=(\bar{A}+B+\bar{C}+D)(A+\bar{B}+\bar{C}+D)(A+\bar{B}+\bar{C}+\bar{D})(A+B+\bar{C}+\bar{D})$

$(A+\bar{B}+\bar{C}+\bar{D})(A+\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+C+\bar{D})$

$=(\bar{A}_1+B_0+\bar{C}_1+D_0)(A_0+\bar{B}_1+\bar{C}_1+D_0)(A_0+\bar{B}_1+\bar{C}_1+\bar{D}_1)$

$(A_0+B_0+\bar{C}_1+\bar{D}_1)(A_0+\bar{B}_1+C_0+\bar{D}_1)(\bar{A}_1+\bar{B}_1+C_0+\bar{D}_1)$

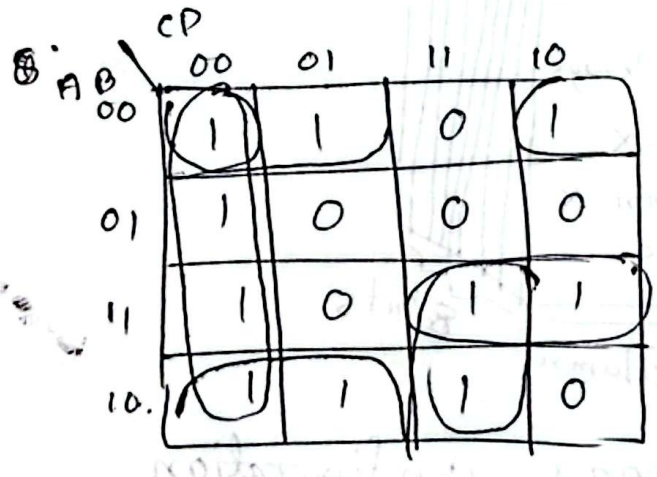
SPOS expression

Truth Table:

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

SSOP:

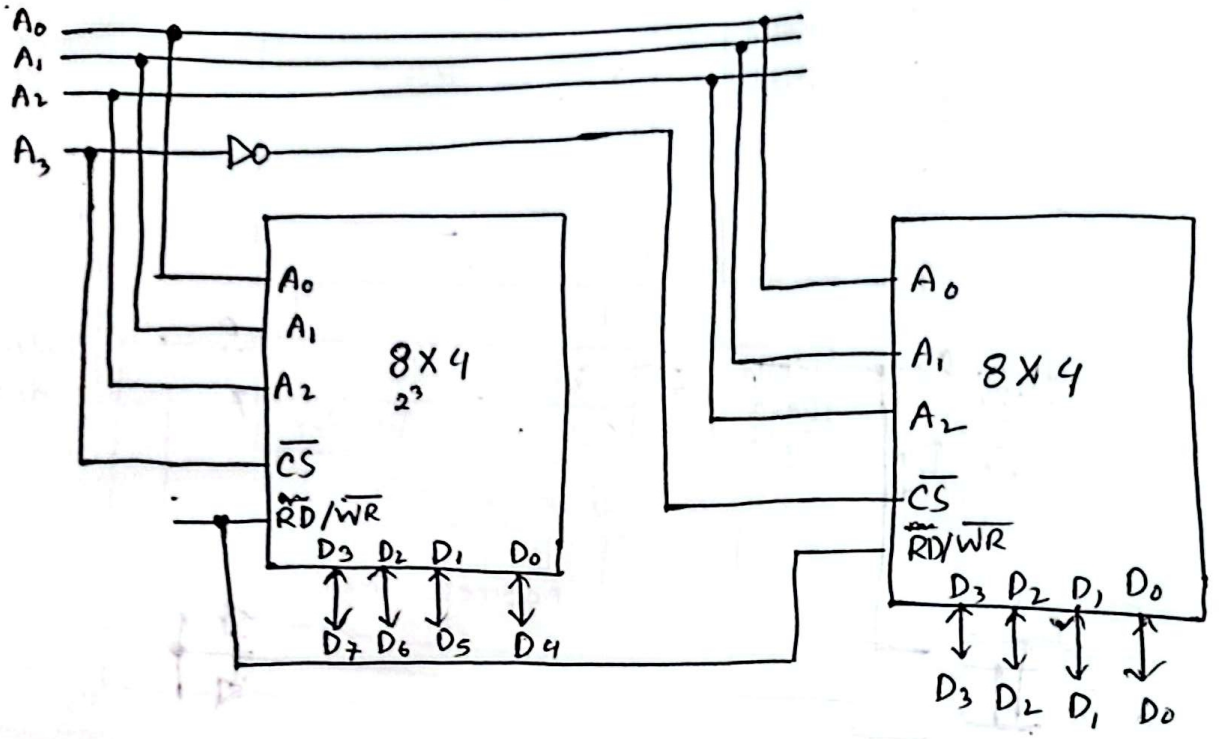
$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABC\bar{D} + ABCD$



Minimum SOP : $\bar{C}\bar{D} + \bar{B}\bar{C} + \bar{A}\bar{B}\bar{D} + ABC + ACD$

5. a)

ii) 16×8 RAM Using 8×4 RAM chip (2^4)



b) i) The R/2R ladder Digital-to-analog converter

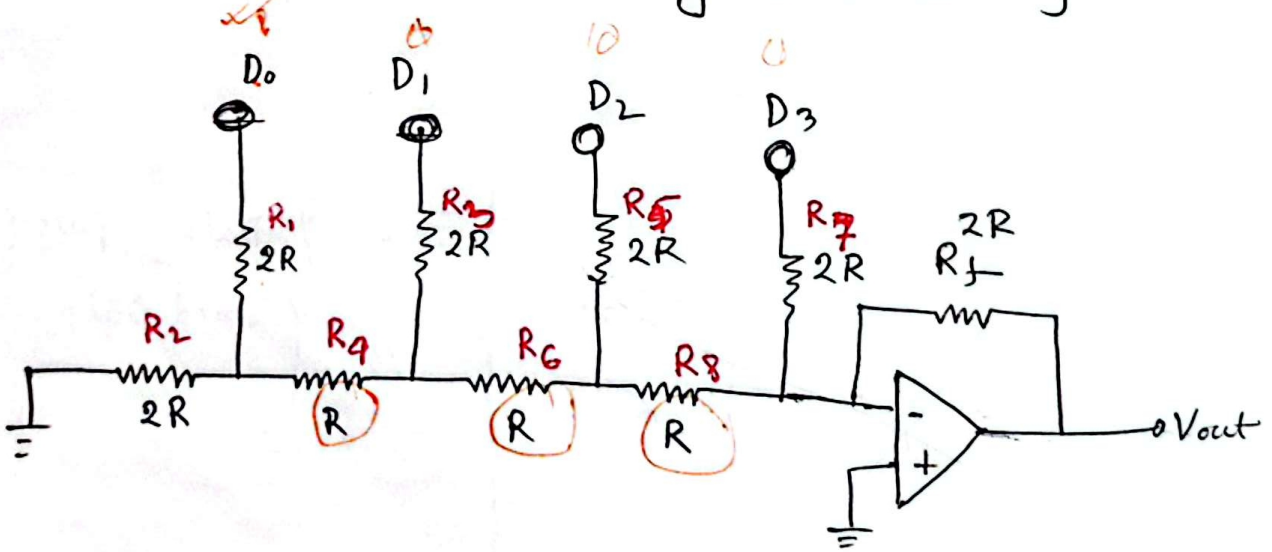


Fig: R/2R Ladder DAC

Operation of R/2R DAC :

1. Resistor network :

→ The core of the DAC is a network of resistors arranged in a repeating pattern. It uses only two resistor values : R and 2R.

→ This consistent pattern is what allows for accurate and scalable conversion.

2. Digital inputs and switches :

→ Each bit of the digital input controls a switch. For bit 1, corresponding switch connects to a reference voltage (V_{ref}) and for bit 0, connects to ground.

3. Voltage division : The R/2R network makes each step's voltage half of the previous one.

4. Output voltages :

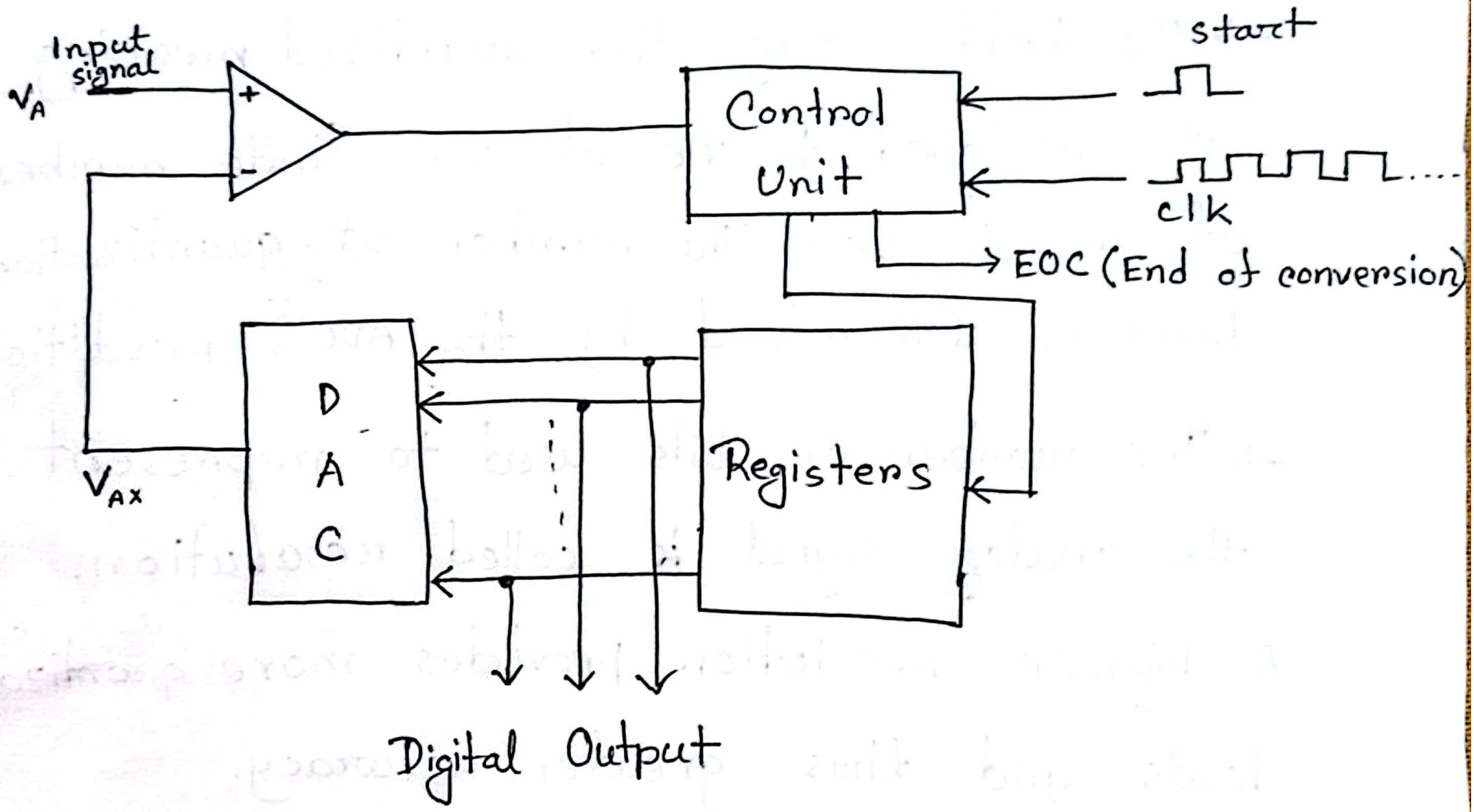
→ The final output voltage is the sum of all the individual voltage contributions from each bit.

→ This results in an analog voltage that is directly proportional to the value of the digital input.

DAC: A DAC converts digital binary data into a proportional analog voltage or current, enabling digital systems to interact with analog devices.

ADC: An ADC takes a continuous analog signal (voltage or current) and converts it into a digital representation (a sequence of binary numbers). This allows computers and digital systems to analyze real world analog signals.

5. c) ii) general ADC diagram :



Working principle:

1. Sampling:

→ The analog input signal is sampled at discrete intervals. The sampling rate is the number of samples taken per second. A higher sampling rate allows for accurate representation of higher frequency signals.

2. Hold:

→ A sample and hold circuit briefly holds the sampled voltage level constants. This ensures that the voltage remains stable during the conversion process.

3. Quantization:

→ The held voltage then quantized, meaning its assigned to one of the finite number of discrete level. The number of quantization level is determined by the ADC's resolution.

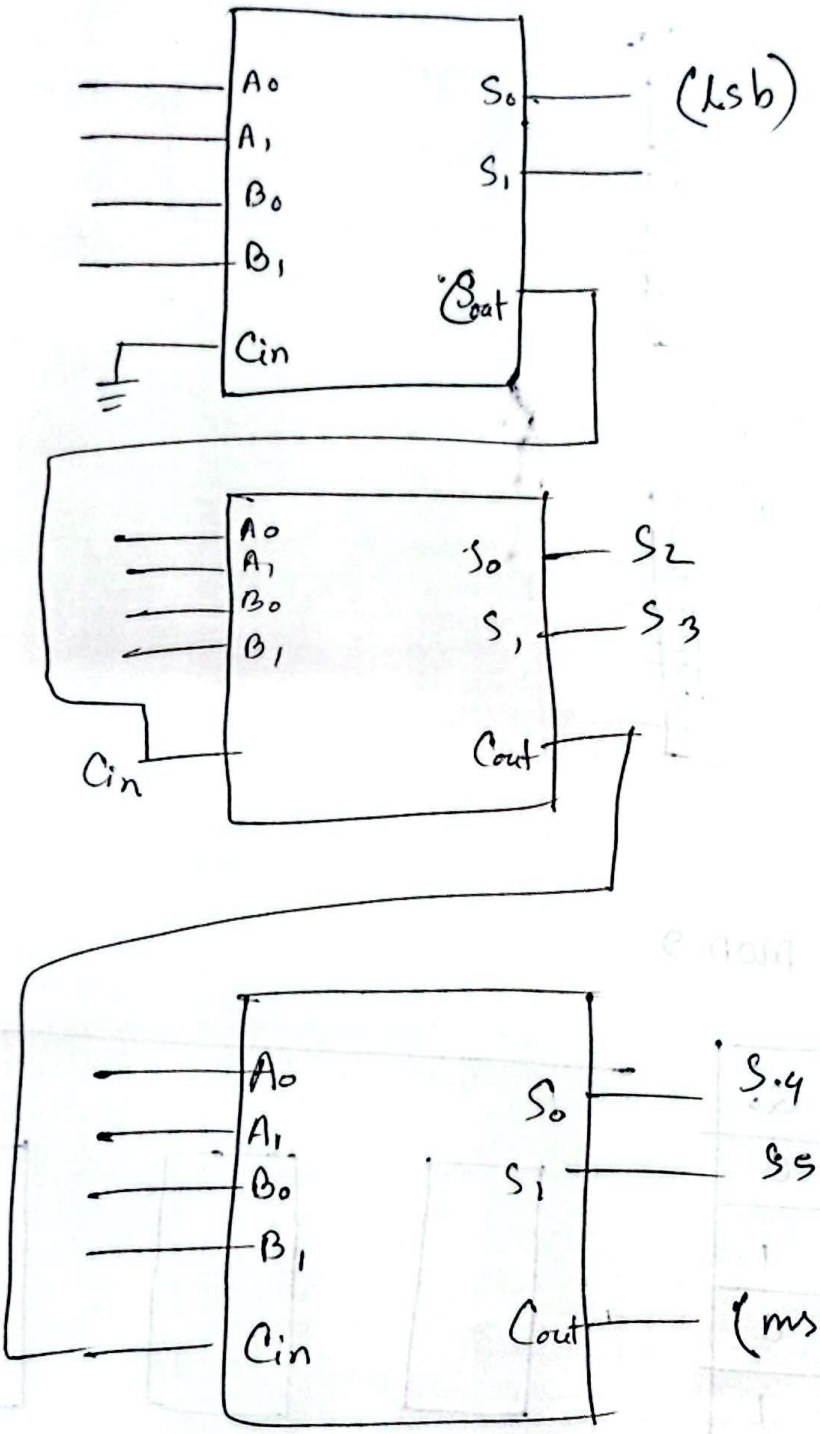
→ The number of bits used to represent the analog signal is called resolution. A higher resolution provides more quantization levels and thus greater accuracy.

→ The difference between the actual analog value and the quantized digital value is called quantization error. This error is inherent in the ADC process.

4. Encoding : Finally the quantized level is encoded into a digital code, typically a binary number.

31 (Tuto - 2)

1. 6 - bit parallel adder from 2 Bit parallel adder



A ⊕ B ⊕ C

AB + (A ⊕ B)Cin

3. 3:8 to 4:16 Decoder

